Saturday 7:38 PM

YOu know we were discussing about a Financial project to select, there was a debate going on to complete it in one month, but the project has to implement somethng, new paper worhty right?

Worked for 7s

Yes. The constraint we had converged on was essentially:

- **~1 month total**
- The project must be **research-oriented**, not just an implementation/demo.
- There needs to be a **specific novelty claim**—something that could plausibly become a paper contribution.
- The baseline should be **reproducible quickly (roughly 5–7 days)**.
- Then the remaining time is spent attacking a **specific limitation/failure boundary** and implementing a genuinely new extension, analysis, or theorem.
- We were explicitly rejecting ideas that merely package known concepts into a new framework.
- In particular, the earlier **finite-sample EMM/scenario-reconstruction** direction was rejected, and the **Heston price-fit vs parameter-recovery** idea was also rejected because the relevant identifiability/conditioning literature was already substantial.

The direction you were favoring was closer to:

> **Take a recent, reproducible quantitative-finance paper/claim → reproduce its result → identify a precise weakness or unexplored boundary → implement a targeted extension → demonstrate a measurable improvement or failure mode.**

So yes, **"finish an interesting financial project in one month" isn't sufficient**. The one-month constraint has to be compatible with a *paper-worthy contribution*.

For the next project selection, I would therefore apply a hard filter:

Requirement	Must satisfy
Baseline reproducible	**≤ 5–7 days**
Existing paper/baseline	**Yes**
Dataset/code available	Preferably **yes**
Novel contribution	**Specific and defensible**
Implementation	**Required**
Quantitative evaluation	**Required**
Novel theorem/algorithm/empirical finding	**At least one**
Can produce meaningful result in ~3 weeks after baseline	**Yes**
"Just apply ML to finance dataset"	**Reject**
"Combine existing methods" without new finding	**Reject**
Entirely theoretical framework requiring months	**Reject**

The important distinction is **paper-worthy ≠ complicated**. A one-month project can be publishable if the research question is sufficiently narrow and the contribution is sharp—for example, discovering that a published numerical method **fails under a particular regime**, proving/characterizing the failure mechanism, and implementing a modification that fixes it.

That is the standard I would use to judge the financial project options we're considering.

This is the course syllabus of that project:

IE612

Introduction to Financial Engineering

Industrial Engineering and Operations Research • 6 credits • Interdisciplinary STEM Course.

The aim of this course is to cover some basic concepts of financial engineering: the issues that arise in modeling, analysis and decision making involving financial instruments. Discrete time models and computational tools will be the focus. Portfolio optimization: Markowitz model; Two and one fund theorems; mutual funds. Capital Asset Pricing model; Security market line. Arbitrage; Hedging; Pricing. Contingent claims; Forward and futures contracts. European and American options; Asian and other path dependent options. One and multi-period binomial models; Finite state models. Equivalent martingale measures; Completeness of markets; Fundamental asset pricing theorems; Option pricing. Black-Scholes option pricing formula.

Just remember this things, I'll tell what to do

Memory updated

Understood. I'll use this IE612 syllabus as the **scope/reference point** for the project discussion.

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Now you know myself, what I know: This is me, more or less most things are included. So this is my technical knowledge, remember that

Memory updated

I'll treat the uploaded resume as the current reference for your technical background, especially when evaluating whether an IE612 project is feasible within your existing skill set.

Your finance/quant background in particular includes **alpha research, financial ML, portfolio optimization, derivatives pricing, Black–Scholes, risk-neutral/martingale pricing, hedging, finite-difference methods, Vasicek interest-rate modeling, and options simulation**.

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I'll also account for your broader **ML, numerical methods, optimization, computational physics, and engineering/control** background when assessing project ideas. PDF bb3be664-5316-4806-82d6-7c0119d... PDF bb3be664-5316-4806-82d6-7c0119d...

Now time for project selection, I have to find resources to decide what project to build for that help find me resources

Yes. Given the **IE612 syllabus + your existing technical background + the one-month constraint + requirement for something genuinely research/paper-worthy**, I would not start by searching "financial project ideas." That produces mostly portfolio/demo projects.

Instead, use a **research-discovery stack**: find recent papers → identify reproducible baselines → inspect limitations/open questions → test whether a narrow extension can be implemented in ~1 month.

1. Start with these research-paper sources

A. arXiv — Quantitative Finance

arXiv Quantitative Finance ↗

This should be one of your primary sources. Look specifically at:

- `q-fin.PR` — Pricing of Securities
- `q-fin.PM` — Portfolio Management
- `q-fin.MF` — Mathematical Finance
- `q-fin.ST` — Statistical Finance
- `q-fin.TR` — Trading and Market Microstructure
- `q-fin.RM` — Risk Management

For your project, don't read papers indiscriminately. Search within these categories for terms such as:

> option pricing, American options, implied volatility, hedging, binomial model, portfolio optimization, robust portfolio, transaction costs, martingale, incomplete markets, regime switching, stochastic volatility, volatility forecasting, reinforcement learning.

There are already recent papers connecting several of these areas to ML/RL. For example, a 2024 paper studies RL for optimal stopping with applications to American option pricing and portfolio choice. arXiv

---

B. SSRN — Financial Economics / Finance

SSRN Financial Economics Research Network ↗

SSRN is particularly useful because finance research often appears there **before journal publication**.

Look for:

- asset pricing
- derivatives
- portfolio management
- financial econometrics
- quantitative finance
- market microstructure

For example, recent work has explored multivariate affine-GARCH models with heavy-tailed innovations for **both portfolio optimization and option valuation**. SSRN

That kind of paper is useful to us not necessarily because we would reproduce it, but because its methodology can reveal **smaller unresolved subproblems**.

C. Google Scholar

Google Scholar ↗

This is probably the most important tool for determining whether an idea is actually novel.

Use searches like:

```
Plain text

"American option" "machine learning" pricing
"binomial tree" option pricing improvement
"robust portfolio optimization" transaction costs
"portfolio optimization" regime switching
"implied volatility" portfolio optimization
"option hedging" machine learning
"equivalent martingale measure" numerical
"market completeness" computational
```

Then filter to **2021–2026**.

For every promising paper, inspect:

1. **References**
2. **Cited by**
3. **Related articles**
4. **Future work**
5. **Limitations**

The last two are particularly valuable for project selection.

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2. Use RePEc/IDEAS for finance literature

IDEAS/RePEc ↗

This is particularly useful for economics/finance because you can search papers by subject and author and follow citation networks.

For example, there is already recent research on ML-based option pricing using Indian NIFTY 50 data, comparing models such as SGD, decision trees, random forests, XGBoost and neural networks against Black-Scholes. IDEAS/RePEc

That immediately tells us something important:

> **"Use ML to improve Black-Scholes pricing" is not, by itself, a novel project idea.**

We would need a much sharper research question.

3. GitHub is extremely important for our one-month constraint

The objective isn't merely to find papers. We want:

> **Paper + code + data + reproducible baseline**

because that can save ~1–2 weeks.

Microsoft Qlib

Microsoft Qlib GitHub ↗

Qlib provides an end-to-end quantitative research platform covering data, models, training and evaluation. GitHub +1

Its documentation is particularly useful for understanding what a modern quant research pipeline looks like. GitHub

However, I would **not automatically choose a Qlib-based project**. It can push us toward generic ML trading projects, which overlap heavily with what you've already done.

Quantopian Alphalens

Alphalens GitHub ↗

Useful for factor/alpha research.

It provides:

- returns analysis
- information coefficient analysis
- turnover analysis
- grouped analysis GitHub

Again, because you already have substantial alpha research experience, I'd use this primarily as a **research infrastructure/reference**, rather than make another generic alpha project.

Zipline / Zipline Reloaded

Zipline GitHub ↗

Useful for understanding rigorous backtesting infrastructure, including event-driven simulation, slippage, transaction costs and avoiding look-ahead bias. GitHub +1

There is also:

Zipline Reloaded ↗

This is more useful than the original Zipline if we actually need a maintained implementation.

---

4. Look for existing research implementations

There are some repositories that are useful specifically as **baseline mines**.

For example:

Quant Option Research Platform ↗

It contains implementations around:

- Black-Scholes
- Greeks
- implied volatility
- Monte Carlo
- volatility modelling
- backtesting

- regime-aware ML
- walk-forward testing
- calibration
- uncertainty analysis GitHub

This is exactly the sort of repository we can inspect to see:

> "What can already be implemented in a few days, and what research question remains beyond it?"

Another useful example:

Monte Carlo Option Pricing Lab ↗

It already covers:

- Monte Carlo pricing
- variance reduction
- American options
- Longstaff-Schwartz
- Greeks
- multi-asset derivatives
- VaR

and explicitly includes validation against Black-Scholes and convergence tests. GitHub

That gives us a potential **baseline**, rather than something we would submit unchanged.

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5. Use current university research pages for project ideas

This is underrated.

University research/thesis pages frequently reveal **problems researchers actually consider open enough to investigate**.

For example, KIT currently lists research around:

> **Portfolio choice with option-implied information using machine learning**

where ML is used to forecast option returns and incorporate option-implied information into mean-variance portfolio construction. C-Financial Econ...

That is much more useful for us than a random "50 finance project ideas" webpage.

Search university pages for:

Plain text

```
site:edu "financial engineering" "research project"
site:edu "option pricing" "machine learning" thesis
site:edu "portfolio optimization" "machine learning" thesis
site:edu "quantitative finance" "project"
site:edu "derivatives" "research project"
```

---

6. Use recent papers to identify the research frontier

A few examples show the kind of territory worth investigating.

American options + RL

A 2024 paper studies learning optimal stopping for diffusion processes and applies the method to American put pricing and portfolio choice.
arXiv

Potential research direction:

> Not "use RL for American options."

Instead:

> **Under what conditions does an RL optimal-stopping method outperform classical Longstaff-Schwartz/binomial approaches, and where does it fail?**

That becomes an empirical research question.

Computational acceleration

A 2025 paper proposes RL to dynamically choose block-preconditioner sizes for iterative solvers appearing in portfolio optimization and option pricing. arXiv

Interesting because it combines:

Financial engineering + numerical optimization + ML

which overlaps unusually well with your background.

Again, though, we would have to check whether there is a sufficiently narrow unexplored extension.

Stochastic volatility / portfolio / pricing

Recent research is combining heavy-tailed volatility models with portfolio optimization and option pricing. SSRN

Potentially interesting, but probably more mathematically demanding than what I'd initially target for a one-month project.

---

7. A very useful source for identifying what actually matters in quantitative research

Campbell Harvey's Finance 656 materials ↗

His current 2026 quantitative asset-management course covers topics including:

- risk
- systematic risk
- strategy evaluation
- rebalancing
- risk targeting
- drawdown control
- crisis hedging
- uncertainty
- factors
- distinguishing luck from skill
- AI/ML in asset management
- quantitative stock selection Duke People

His research page is also useful for following the literature:

Campbell Harvey Research Papers ↗ Duke People

This is particularly useful for **research methodology**, not just project ideas.

---

8. Don't ignore textbooks

For IE612 specifically, I would keep one strong reference open while exploring papers.

Hull — Options, Futures, and Other Derivatives

Useful for:

- binomial trees
- Black-Scholes
- futures/forwards

- Greeks
- hedging
- American options
- exotic options

Shreve — Stochastic Calculus for Finance

Useful when the project gets into:

- martingales
- risk-neutral measures
- arbitrage
- stochastic processes
- option pricing

Glasserman — Monte Carlo Methods in Financial Engineering

Especially relevant if we go toward:

- Monte Carlo pricing
- variance reduction
- Greeks
- rare events
- computational acceleration

These shouldn't be our **idea generators**. They should be our **theoretical verification layer**.

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9. One resource I would particularly use for your project

Because you've already done substantial quant work, I would use **WorldQuant's research material only as a secondary resource**:

WorldQuant Learn2Quant ↗

It covers alpha construction, diversification, risk management and advanced quantitative research techniques. WorldQuant

But I would deliberately avoid letting this turn the IE612 project into another **alpha-generation/backtesting project**. You already have much stronger experience there, and it doesn't naturally exploit the full IE612 syllabus.

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10. The research-selection pipeline I'd use

Rather than searching randomly, I'd make a spreadsheet with these columns:

Paper	Year	IE612 topic	Baseline available?	Code?	Data?	Main limitation	Possible extension	Imp
Paper A	2025	American options	✓	✓	✓	X	Y	
Paper B	2024	Portfolio	✓	X	✓	X	Y	
Paper C	2026	Hedging	✓	✓	X	X	Y	

And I would **not select the project until we have at least ~10–20 candidate papers**.

Then eliminate anything where:

Reject immediately

- It is simply an implementation of a textbook model.
- "Compare XGBoost vs Random Forest vs neural network."
- "Apply deep learning to option pricing."
- "Use RL for portfolio optimization."
- The novelty is merely combining two existing techniques.
- It requires proprietary data.

- The baseline itself takes >1 week.
- The paper's claimed contribution is already extensively extended by later papers.
- The only contribution would be testing it on Indian data.
- It requires a mathematical derivation substantially beyond your available one-month time.

Keep for deeper investigation

- A paper has a **clear limitation**.
- There is an identifiable unexplored parameter/regime/boundary.
- A theoretical result has an interesting empirical consequence we can test.
- A numerical method has a known computational weakness.
- A model makes an assumption that can be systematically relaxed.
- A method works under one market condition but its **failure boundary isn't characterized**.
- There is a reproducible baseline.
- We can make the new experiment in **2–3 weeks**.

The three research areas I'd investigate first for you

Given **your background + IE612 + one month**, I would search these three areas first:

1. Derivative pricing / hedging + numerical methods

Especially:

Binomial / Monte Carlo / finite difference / American options / Greeks / hedging

This lets you leverage your existing derivatives work while moving beyond the basic Black-Scholes implementation.

2. Portfolio optimization under realistic constraints

Not another Markowitz implementation.

Instead investigate:

transaction costs, estimation error, regime shifts, distributional uncertainty, robust optimization, option-implied information, dynamic rebalancing

This connects directly to IE612's portfolio material and your optimization/ML background.

3. Market completeness / risk-neutral pricing / discrete-time models

This is less crowded with generic ML projects.

For example:

> Can we computationally characterize how approximation/discretization/model misspecification affects the existence or stability of an equivalent martingale measure?

That type of question could potentially produce a much more **financial-engineering-specific** project than another ML prediction exercise.

One important point

I would **not yet choose a project from these resources**.

The next step should be a **literature sweep**, where we systematically collect perhaps **30–50 recent papers (roughly 2021–2026)** across the IE612 topics, then classify each into:

already saturated / reproducible / promising gap / too difficult / too generic / potentially one-month feasible.

From that, we can narrow it to perhaps **5–8 serious project candidates**, and only then investigate novelty deeply.

If you want, I can do that literature sweep next and build the candidate database rather than just giving you more generic resource links.

> I think this is a good project to work: Portfolio choice with option-implied information using machine learning

> We can incorporate Matringale and Makrov models + ML algos to predict option pricing and use that to get ane xepcted stock price under no-arbitrage and use that in protfolio construction

Yes, **the core direction is strong**, but I would change one important part of the proposed architecture.

The project should **not** be framed as:

> ML predicts option prices → use no-arbitrage → obtain expected stock price → Markowitz portfolio.

That has a theoretical problem: **no-arbitrage/risk-neutral pricing does not by itself give the physical expected stock return**. Option prices primarily identify information under the **risk-neutral measure** Q. To turn that into an investment expectation under the physical measure P, you need an additional model for the risk premium / stochastic discount factor / pricing kernel.

That distinction can actually become the **interesting research contribution**.

What the literature says

The basic idea is already established.

A 2020/2021 paper explicitly studies optimal portfolio allocation using option-implied information, constructing state variables for the **risk premium and market price of risk**, and using implied risk-neutral densities. Wiley Online Lib... +1

Earlier work has also used option-implied distributions for portfolio choice, including martingale-based approaches. Aït-Sahalia and Brandt, for example, use option-implied state prices in a portfolio/consumption problem. National Bureau...

And there is already evidence that option-implied variables can predict future equity returns and realized variance. One study finds predictive information in the correlation and variance risk premia and evaluates the economic value for portfolio timing. Taylor & Francis ...

Even more directly, the **exact project title you found is a current KIT research topic**: the proposed project uses ML such as LASSO to forecast option returns and then uses option-implied information for mean-variance portfolio selection. C-Financial Econ...

So the idea is definitely legitimate—but **we cannot claim novelty merely from combining option information + ML + portfolio optimization**.

I would modify your idea into this

Proposed architecture

Options Market

↓

1. Extract option-implied information

From the cross-section of calls/puts:

$$C(K,T), P(K,T)$$

extract:

- implied volatility
- volatility skew
- term structure
- risk-neutral moments
- risk-neutral density
- variance risk premium
- skewness risk premium
- possibly higher moments

↓

2. Model the latent market state

Here your **Markov model** becomes useful.

For example:

$$Z_t \in \{\text{Low Vol}, \text{Normal}, \text{High Vol}\}$$

with

$$P(Z_{t+1} = j | Z_t = i) = P_{ij}$$

using an HMM / Markov regime-switching model.

The state can depend on:

- historical returns
- realized volatility
- option-implied volatility
- skew
- risk-neutral moments
- macro variables

This gives:

$$P(Z_{t+1} | I_t)$$

rather than simply predicting returns with an ML black box.

---

3. ML predicts the physical distribution

This is where I think your idea can become substantially more interesting.

Instead of:

$$ML \to E[S_{t+1}]$$

we attempt:

$$ML: \quad I_t \to P\text{-distribution of } R_{t+1}$$

or at least:

$$\hat{\mu}_{t+1}, \quad \hat{\sigma}^2_{t+1}, \quad \hat{\text{Skew}}_{t+1}, \quad \hat{\text{Kurtosis}}_{t+1}$$

The input contains **option-implied information**, not just historical stock returns.

For example:

$$X_t = [IV_t, Skew_t, TermStructure_t, VRP_t, RRP_t, R_t, RV_t, Macro_t]$$

and ML predicts:

$$\hat{\mu}_{t+1} = f_\theta(X_t)$$

---

4. The critical theoretical bridge: $Q \to P$

This is where **martingale theory** becomes genuinely relevant.

Under the risk-neutral measure:

$$E^Q_t \left[\frac{S_{t+1}}{B_{t+1}} \right] = \frac{S_t}{B_t}$$

but the investor cares about:

$$E^P_t [R_{t+1}]$$

Those aren't automatically equal.

Conceptually:

$$\boxed{Q\text{-distribution} \neq P\text{-distribution}}$$

The difference contains information about **risk premia**.

One possible framework is a stochastic discount factor:

$$M_{t+1} = \frac{dQ}{dP}$$

and

$$E_t^P[M_{t+1}R_{t+1}] = 0$$

for excess returns.

This gives us a principled way of asking:

> **Can ML learn the transformation from option-implied risk-neutral information into economically useful physical-measure forecasts while respecting no-arbitrage restrictions?**

That's considerably more interesting than "ML predicts options."

---

5. Then portfolio construction

Once we have:

$$\hat{\mu}_{t+1}$$

and

$$\hat{\Sigma}_{t+1}$$

we solve something like:

$$\max_{w} \quad w^\top \hat{\mu} - \frac{\lambda}{2} w^\top \hat{\Sigma} w$$

subject to:

$$\sum_i w_i = 1$$

and potentially:

$$w_i \geq 0$$

plus:

- turnover constraints
- transaction costs
- position limits
- volatility target

Then compare:

Baseline 1

Historical Markowitz

$$(\mu_{historical}, \Sigma_{historical})$$

Baseline 2

Option-implied portfolio

$$(\mu_{OI}, \Sigma_{OI})$$

Baseline 3

ML using historical information

$$(\mu_{ML}, \Sigma_{ML})$$

Proposed model

$$\boxed{\text{Option information} + \text{Markov state} + \text{ML} + \text{no-arbitrage constraints}}$$

Where I think the real research question should be

Rather than claiming:

> "We build an ML portfolio using options."

I'd formulate the research question approximately as:

> **Can option-implied risk-neutral information, combined with regime-dependent machine learning estimates of the physical return distribution, improve out-of-sample portfolio allocation relative to historical and purely option-implied approaches?**

And then an even more interesting sub-question:

> **Does explicitly modelling the Q-to-P transformation through regime-dependent risk premia provide incremental portfolio value over directly predicting returns from option characteristics?**

That gives us an actual experimental hypothesis.

And your IE612 syllabus fits this unusually well

IE612 concept	Project component
Markowitz	Portfolio optimization
CAPM	Expected return / risk premium baseline
Arbitrage	No-arbitrage constraints
Hedging	Optional extension / validation
Contingent claims	Options
European options	Option data
American options	Potential extension
Binomial models	Risk-neutral distribution baseline
Finite-state models	Markov regime model
Equivalent martingale measures	Q-measure framework
Completeness	Limitations of recovering distributions
Fundamental asset-pricing theorem	No-arbitrage foundation
Black-Scholes	Benchmark implied-volatility extraction

So the project can remain **inside IE612**, while using your ML/quant background to go beyond the basic syllabus.

But there is a major novelty problem we need to solve

There is already substantial literature on:

- option-implied information → portfolio allocation Wiley Online Lib...
- option-implied variables → return prediction Taylor & Francis ...
- option-implied volatility/skewness → portfolio selection Cambridge Univ...
- risk-adjusted option-implied distributions → portfolio choice PubsOnLine
- ML → option-implied information extraction Taylor & Francis ...
- ML/high-dimensional methods → cross-sectional stock-return prediction from option characteristics; a 2025 *Management Science* paper explicitly studies this. Google Sites

And the KIT project itself proposes **LASSO + option returns + portfolio selection**. C-Financial Econ...

Therefore, **we should not yet commit to the exact architecture you proposed.**

Instead, I think the next research task should be:

Literature-gap investigation

Take these four components separately:

$$\boxed{\text{Option-implied information}}$$

$$\boxed{Q \to P \text{ transformation}}$$

$$\boxed{\text{Markov/regime switching}}$$

$$\boxed{\text{ML portfolio optimization}}$$

and search **2020–2026 literature specifically for papers combining each pair/triple**.

The goal is to find a hole such as:

option-implied → portfolio: already done
option-implied → ML prediction: already done
Markov → option pricing: already done
ML → portfolio: heavily done
option-implied + regime-dependent $Q \to P$ transformation + portfolio construction: potentially less explored

If that gap survives a proper literature search, **then we have a potentially very good one-month IE612 project**.

And importantly, we can make the first version much smaller: **S&P 500 constituents / liquid options → option-implied moments → regime model → constrained ML $Q \to P$ forecast → Markowitz → strict walk-forward evaluation.** That is computationally realistic for your existing skill set.

> Good, we will go in these direction, your role is to just find 3-4 resrearch papers through which I can understand how to start the proejct, see use research papes levels I can understand, you know me

Yes. Given what you already know, I would **not** give you introductory finance papers or highly mathematical stochastic-calculus papers. You need a sequence where each paper teaches one component of the eventual project.

I would read these **4 papers in this order**:

1. DeMiguel et al. (2013) — start here

"Improving Portfolio Selection Using Option-Implied Volatility and Skewness"
Journal of Financial and Quantitative Analysis

Paper / SSRN full text ↗

This is the **best starting point for your project**.

The paper asks almost exactly the first-level question we care about:

> Can option-implied information improve mean-variance portfolio selection?

They examine option-implied:

- volatility
- correlation
- risk premium
- skewness

and incorporate this information into expected-return/portfolio estimation. They evaluate out-of-sample volatility, Sharpe ratio and turnover.

Cambridge Univ... +1

What you should learn from it:

```
Option prices
     ↓
Option-implied quantities
     ↓
Expected return / risk estimates
     ↓
Mean-variance portfolio
     ↓
Out-of-sample evaluation
```

This gives you the **base architecture** of our project.

It is also at a level you should be able to follow without learning a huge amount of new mathematics.

---

2. Bianchi & Tassinari (2018) — learn the Q → P bridge

"Forward-looking portfolio selection with multivariate non-Gaussian models and the Esscher transform"

arXiv paper ↗

This one is much closer to the part **you specifically want to add**.

The authors construct a portfolio framework using **option-implied information**, while modelling non-Gaussian return distributions. Crucially, they connect the historical/physical measure P and risk-neutral measure Q using the **Esscher transform**. arXiv

The basic idea is:

$$P \quad \longrightarrow \quad \text{return dynamics}$$

and

$$Q \quad \longrightarrow \quad \text{option prices}$$

with a transformation connecting them.

They then use the resulting distributional information for portfolio selection.

This is the paper I want you to pay particular attention to, because it prevents us from making the mistake we discussed earlier:

$$\boxed{\text{risk-neutral expected price} \neq \text{physical expected price}}$$

You need some mechanism to bridge Q and P.

For you, this paper is probably the conceptual bridge between **IE612's equivalent martingale measures** and the portfolio/ML project.

---

3. Chernov & Ghysels (2000) — understand how option data informs P and Q

"A Study Towards a Unified Approach to the Joint Estimation of Objective and Risk Neutral Measures for the Purpose of Options Valuation"

Journal of Financial Economics paper ↗

Don't try to reproduce the mathematics.

Read it for the **conceptual architecture**.

The paper explicitly tries to bridge:

$$\boxed{\text{physical/objective measure } P}$$

and

$$\boxed{\text{risk-neutral measure } Q}$$

using both the underlying asset and option information. They demonstrate the approach using Heston's model and S&P 500 data. ScienceDirect +1

This gives you the theoretical background for the question:

What exactly are we learning from the options market that isn't already contained in historical stock returns?

That is a very important question for our project.

---

4. Gu et al. (2023) — then bring in ML

"Option Return Predictability with Machine Learning and Big Data"
Review of Financial Studies

Oxford Academic paper ↗

This is where your existing ML/quant knowledge becomes directly relevant.

The paper uses **12+ million observations from 1996–2020** and investigates nonlinear ML relationships between option/stock characteristics and future option returns. It finds that nonlinear ML models provide substantial out-of-sample predictive information, with option characteristics being particularly important predictors. OUP Academic

You don't need to learn ML from this paper—you already know that part.

Instead, study:

- What are their input features?
- What exactly is the prediction target?
- How do they construct the training/test split?
- How do they avoid look-ahead bias?
- What constitutes an economically meaningful prediction?
- How do they convert predictions into portfolio positions?
- How do they evaluate out-of-sample performance?

Those are the **research-design lessons** we need.

So these four papers form a progression

I would visualize your eventual project as:

Plain text

And each paper teaches one block:

Paper	What you learn
DeMiguel et al. 2013	Option information → portfolio optimization
Bianchi & Tassinari 2018	$P \leftrightarrow Q$ transformation → portfolio choice
Chernov & Ghysels 2000	Why/how physical and risk-neutral measures are jointly connected
Gu et al. 2023	ML + option information + rigorous out-of-sample prediction

One newer paper is worth keeping in reserve

There is a **2026 paper** called *"Forward-looking physical tail risk: a deep learning approach"* that is extremely close to the direction we're discussing. It explicitly tries to extract **physical-measure information from risk-neutral information using deep learning**, using S&P 500 option data.

Taylor & Francis ... +1

I would **not start with this one**.

It is closer to the *research frontier* and could be very useful later for identifying what has already been done. In particular, its existence means we have to be careful about claiming that simply doing

$$Q \to \mathrm{ML} \to P$$

is novel.

Your reading order

Read only the abstracts/introduction/conclusion first:

DeMiguel → Bianchi → Chernov/Ghysels → Gu et al.

Then, on the second pass, read the **methodology sections** of DeMiguel and Bianchi carefully.

After that, we can take these four papers and construct a **"what already exists → what each paper leaves open → what our one-month contribution could actually be"** map. That is the point at which I would start designing the actual project rather than adding more components blindly.

Today 1:22 AM

From any one of this, would I be able to get the options data for US or Indian Market?

Databases / Standard Databases

- **ABI/Inform Database**
- **American Ceramic Society-Phase Equilibria Diagrams** (10 Users) (Registration required)
- **Bloomberg** (Terminal available at Central Library, Circulation Area)
- **Bloomsbury Visual Arts Hub**
- **Crisil Research**
- **Capitaline AWS Database** (To use this facility please register using IITB email id)
- **CMIE Database**
- **Dataful** (Click on "Sign In" and then click on "Sign Up with IP")
- **EMIS**
- **EPWRF India Time Series**
- **Euromonitor Passport**
- **Grove Music Library**
- **HS Talks Business and Management Collection**
- **Indian Boards Database**
- **ICSD : Inorganic Crystal Structure Database**
- **IndiaStat** (Click IP Login)
- **Inspec Database**
- **Journal Citation Report**

- **LSEG Workspace** (Previously known as Refinitiv : For Login contact Journals Section)
- **MathSciNet**
- **Naxos Music Library** (Contact: Journal Section)
- **ProQuest**
- **Qualtrics** (For Login contact Journals Section)
- **Reaxys Database**
- **SciFinder Scholar**
- **Scite (Smart Citation)**
- **SCOPUS**
- **Statista**
- **The Cambridge Structural Database (CSD)** (For Activation key contact Journal Section)
- **The Ken** (For Registration)
- **The Ken** (For Login)
- **Tracxn Database** **(User Guide)**
- **VCCEdge Database** (Contact: Journal Section)
- **Venture Intelligence Database**
- **Web of Science**
- **YNOS Venture Engine**

Standard Databases

- **ASTM Digital Library With Standards**
- **IEC Standards**
- **Indian Standards**
- **ISO Standard (Custom Collection)**

These and also any open source like yahoo and all for this particular project?

Yes. For **this specific project**, your IIT Bombay database access is potentially very useful. In fact, I would **prefer institutional Bloomberg/LSEG data over trying to build the entire project from Yahoo Finance

From any one of this, would I be able to get the options data for US or Indian Market?

Databases / Standard Databases

- **ABI/Inform Database**
- **Bloomsbury Visual Arts Hub**
- **Crisil Research**
- **Capitaline AWS Database** (To use this facility please register using IITB email id)
- **CMIE Database**
- **Dataful** (Click on “Sign In” and then click on “Sign Up with IP”)
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Standard Databases

- **ASTM Digital Library With Standards**
- **IEC Standards**
- **Indian Standards**
- **ISO Standard (Custom Collection)**

These and also any open source like yahoo and all for this particular project?

Yes. I checked the databases you listed specifically against the data requirements of **our proposed project**.

The important distinction is that we don't merely need "stock prices." We need an **option panel over time**:

$$(t,\ S_t,\ K,\ T,\ C/P,\ P_t,\ Volume,\ OI,\ IV, \ldots)$$

ideally including **expired contracts**, because otherwise we cannot do a proper historical walk-forward study.

Short answer

Source	US options	Indian options	Historical options	Useful for our project?
LSEG Workspace	**Yes**	**Yes, depending on subscribed exchange package**	**Yes, potentially extensive**	⭐⭐⭐⭐⭐
Bloomberg Terminal	**Yes**	**Yes**	**Yes, depending on terminal/data entitlement**	⭐⭐⭐⭐⭐
NSE	No	**Yes**	**Yes**	⭐⭐⭐⭐
Capitaline	No	Some derivatives/market data	Need to verify exact options history	⭐⭐⭐
CMIE Prowess	No	**Underlying equities, fundamentals**	Yes	⭐⭐
Dataful	Primarily India datasets	Possibly derivatives-related datasets	Depends on dataset	⭐⭐
Yahoo Finance	**Current option chains**	Limited/poor for Indian options	**Not sufficient for historical option panel**	⭐
Other open-source sources	Sometimes	Sometimes	Usually fragmented	⭐–⭐⭐

The two sources I would investigate **first at IIT Bombay are LSEG and Bloomberg**.

---

1. LSEG Workspace — probably your best option

This is the most interesting one on your list.

LSEG explicitly provides **exchange-traded options data**, including:

- US options
- international options
- option reference data
- historical contracts
- pricing
- implied volatility

- Greeks
- volatility surfaces

LSEG says its derivatives coverage spans **100+ exchanges**, and its options reference data includes both active and expired options. Its stated history includes selected contracts from 2000 and higher-quality options histories from 2018 depending on the product. LSEG +1

More importantly, LSEG explicitly lists **OPRA US options market data** among its derivatives datasets. LSEG

So conceptually:

LSEG

could give us:

```
Plain text

US:
S&P 500
SPY
individual equities
options
strikes
expiries
prices
volume
OI
IV
Greeks
...
```

and potentially Indian exchange derivatives as well.

BUT

This is the critical caveat:

I cannot conclude from IITB's library listing alone that your particular IITB LSEG Workspace subscription includes historical US/Indian options pricing.

IITB has access to **LSEG Workspace**, but the exact data entitlements are subscription-dependent.

So this is the first thing I would check with the library/data desk.

Ask them specifically:

> "Does IIT Bombay's LSEG Workspace subscription include historical exchange-traded equity/index options data for US markets (OPRA/US options) and NSE India? In particular, can we download historical option contract-level data containing trade date, expiration, strike, call/put, price, volume and open interest?"

That wording matters.

---

2. Bloomberg Terminal — also potentially excellent

You have:

> **Bloomberg — Terminal available at Central Library, Circulation Area**

That is potentially very useful.

Bloomberg provides options information through the terminal, including option chains and derivatives analytics.

For our project, however, don't just ask:

> "Can Bloomberg show options?"

Obviously it can.

Ask:

Can I download historical option-chain/contract-level data for expired US equity/index options and/or NSE options, including strike, expiration, call/put, price, volume and open interest?

Because our project requires **historical reconstruction**, not simply looking at today's chain.

US

Bloomberg should be particularly attractive for:

$$S\&P500/SPY/individual\ US\ stocks$$

because the US options market is enormously deep.

India

Potentially:

$$NIFTY,\ BANKNIFTY,\ \text{NSE stocks}$$

but again, **verify the exact historical entitlement**.

---

3. NSE itself — surprisingly useful for India

If we decide to do the project on India, NSE is actually a very interesting source.

The NSE option chain exposes:

- strike
- call/put
- LTP
- bid
- ask
- volume
- open interest
- implied volatility

for equity derivatives. NSE India +1

NSE also provides **historical contract-wise price-volume data** with fields including:

- date
- expiry
- open/high/low/close
- LTP
- settlement price
- number of contracts
- turnover
- open interest
- change in OI
- underlying value. NSE India

And NSE explicitly provides historical F&O data through its paid historical-data service, including F&O order/trade data. NSE India

So there is a possible India pipeline:

$$\boxed{\text{NSE options}} \rightarrow \text{IV / risk-neutral distribution} \rightarrow \text{ML} \rightarrow \text{portfolio}$$

However, there is a problem.

The freely accessible NSE historical contract interface currently restricts the selectable time period to **90 days**. NSE India

So for a **multi-year academic dataset**, we'd need either:

1. IITB's licensed data source, or
2. NSE's paid historical dataset, or
3. another provider.

Also, NSE's website explicitly places restrictions on copying/aggregating data from the site, so I would **not build a research pipeline by scraping the live option-chain webpage**. NSE India

---

4. Capitaline

Capitaline is definitely useful for the **Indian side of the project**, but I wouldn't rely on it yet for options.

Capitaline says its database includes market data on Indian securities, derivatives and commodities. Capitaline

However, what I can verify publicly is much stronger for:

- company fundamentals
- equity prices
- financial statements
- ratios
- corporate data

than for the **full historical option-contract panel** we need.

So:

$$\text{Capitaline} \rightarrow \boxed{\text{Excellent supplementary Indian equity/fundamental data}}$$

but:

$$\text{Capitaline} \not\Rightarrow \boxed{\text{confirmed historical options dataset}}$$

until we inspect your IITB subscription/interface.

---

5. CMIE Prowess — useful, but NOT for options

CMIE Prowess is primarily an Indian company database.

It has:

- financial statements
- quarterly data
- daily share prices
- corporate actions
- total returns
- company characteristics

and long historical coverage. CMIE +1

So it could supply:

$$\text{Indian stock fundamentals} + \text{historical stock returns}$$

but I would **not expect it to supply the option chain required by our project**.

It could nevertheless become useful later for additional ML features.

---

6. Yahoo Finance — don't use it as the main options source

This is where I'd be particularly careful.

Yahoo is fine for:

$$S_t$$

and basic current option-chain information.

But our project requires something much more demanding:

$$\{C_{t,K,T}, P_{t,K,T}\}_{t=1}^{T}$$

across many historical dates.

Yahoo Finance is **not a sufficiently reliable institutional historical options database for this project**.

So I would use Yahoo for:

- quick prototyping
- underlying prices
- sanity checks

but **not as the research dataset**.

---

7. There is an even more important issue: US vs India

For this project, I would actually lean toward **US data** if LSEG/Bloomberg gives us proper historical access.

Why?

Because our proposed model needs a reasonably rich cross-section:

$$\text{stock} \times \text{date} \times \text{strike} \times \text{maturity}$$

The US options market gives us an enormous cross-section.

That makes it much easier to estimate:

$$\text{Risk-neutral density}$$

and features such as:

$$IV, \quad skew, \quad term\ structure, \quad VRP, \quad risk\ neutral\ moments.$$

For example:

$$\text{SPY options}$$

alone can give us a huge amount of information.

And then we could construct a portfolio of, say:

$$100 - 500$$

liquid US equities depending on data availability.

---

8. The dataset we actually need

Before selecting US vs India, I would define the minimum dataset.

Underlying

For each stock:

```
Plain text
```

```
date
close
adjusted close
returns
volume
```

Option

For each contract:

```
Plain text
```

```
date
underlying
option_type
strike
expiration
bid
ask
mid
last
volume
open_interest
implied_volatility
```

Ideally:

```
delta
gamma
vega
theta
```

although we can calculate these ourselves.

Additional

```
risk_free_rate
dividend_yield
market_return
```

Then:

$$\boxed{\text{Options panel} + \text{Underlying panel} + \text{Risk-free rates}}$$

is enough to start building the research pipeline.

---

9. One particularly important thing for our project

We don't actually need **option prices predicted by ML** as the first step.

I'd revise our project architecture slightly.

We can extract the risk-neutral distribution from observed option prices:

$$C(K,T) \to q_t(S_T)$$

using something like the Breeden-Litzenberger relationship:

$$q(K) \propto \frac{\partial^2 C(K,T)}{\partial K^2}$$

Then:

$$q_t(S_T) \to \text{risk-neutral moments}$$

giving:

$$E^Q[S_T], \quad Var^Q(S_T), \quad Skew^Q(S_T), \dots$$

Then the interesting problem becomes:

$$\boxed{Q\text{-information} \to P\text{-forecast}}$$

using:

- Markov/regime model

- ML
- risk-premium model

and **then** portfolio construction.

That is much more defensible than simply training an ML model to predict option prices.

What I would do at IITB

First priority — LSEG

Go to the LSEG Workspace terminal and determine whether you can access:

US:

- SPX
- SPY
- individual US equity options
- historical expired contracts

India:

- NIFTY
- BANKNIFTY
- NSE equity options
- historical expired contracts

Second priority — Bloomberg

Check the same thing.

Third — NSE

Determine what historical F&O data IITB can legally obtain/download.

Fourth — Yahoo/open-source

Use only for prototyping and cross-checking.

My current ranking for our project

If the data is available through IITB:

$$\boxed{\text{LSEG} \approx \text{Bloomberg} > \text{NSE} > \text{Capitaline/CMIE} > \text{Yahoo}}$$

And I would **prefer US options over Indian options initially**, unless IITB's Indian options history is substantially easier to obtain.

The next thing I would do is **not start coding yet**. First, we should determine exactly what **LSEG Workspace and Bloomberg at IIT Bombay can export**. Once we know that, we can choose US vs India based on the actual available historical option panel rather than guessing.

Bruh forget bloomberg this is the updated list:

From any one of this, would I be able to get the options data for US or Indian Market?

Databases / Standard Databases

- **ABI/Inform Database**
- **Bloomsbury Visual Arts Hub**
- **Crisil Research**
- **Capitaline AWS Database** (To use this facility please register using IITB email id)
- **CMIE Database**
- **Dataful** (Click on “Sign In” and then click on “Sign Up with IP”)
- **EMIS**
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- **Euromonitor Passport**
- **Grove Music Library**

- **HS Talks Business and Management Collection**
- **Indian Boards Database**
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- **Venture Intelligence Database**
- **Web of Science**
- **YNOS Venture Engine**

Standard Databases

- **ASTM Digital Library With Standards**
- **IEC Standards**
- **Indian Standards**
- **ISO Standard (Custom Collection)**

These and also any open source like yahoo and all for this particular project?

Yes — with the **updated list**, the answer is much clearer. I checked the relevant providers rather than assuming that every finance database has options.

Bottom line

For **our project**, among the databases you listed:

🟢 Most promising

Database	US options	Indian options	Historical contract-level options?	Verdict
LSEG Workspace	**YES**	**Potentially YES**	**YES**	**Best option if IITB subscription includes it**
Capitaline	No	**Some derivatives data**	**Possibly, but not confirmed for full option panel**	Useful backup for India
Dataful	Unclear	**Possibly**	Dataset-dependent	Investigate
CMIE/Prowess	No	No options	—	Useful for stock/fundamental data only
EPWRF	No	Primarily macro/financial time series	Not suitable for option chain	Not our options source
CRISIL Research	No	Mostly Indian market/fundamental/credit research	Not a suitable options panel source	Not primary
Everything else	No	No	—	Not relevant

🟡 Open-source

Source	Options	Historical options	Use
Yahoo Finance	US	**Very limited for our purpose**	Prototype/sanity check

Source	Options	Historical options	Use
NSE	Indian	Some historical F&O data	Potentially useful
Stooq / Alpha Vantage etc.	Generally not suitable	No	Don't depend on them
GitHub datasets	Sometimes	Dataset-dependent	Useful if properly documented

---

1. LSEG is the one I would investigate first

Your list has:

> **LSEG Workspace (Previously Refinitiv)**

This is by far the strongest candidate.

LSEG's own documentation confirms that it has **global exchange-traded options data**, including historical options. Their options reference dataset covers **100+ regulated exchanges across 35 countries**, with more than **120 million expired options** where exchange coverage is available. LSEG

For the US specifically, LSEG provides **OPRA** data, which consolidates data from 17 US options exchanges and covers US equity, index and ETF options. It includes:

- last-sale data
- quotes
- NBBO
- contract specifications
- expiration/series information
- historical observations
- analytics/Greeks where available LSEG

That is almost exactly the type of data our project needs.

So:

$$\boxed{\text{US options} \rightarrow \text{LSEG: YES}}$$

The important question is whether **IIT Bombay's particular LSEG subscription gives you access to the relevant historical dataset**.

Don't assume that because IITB has "LSEG Workspace" listed, every LSEG dataset is automatically included.

---

2. What about Indian options through LSEG?

This is also plausible.

LSEG says its derivatives coverage spans global regulated exchanges across Asia-Pacific as well as North America. LSEG

But again, we need to distinguish:

> **LSEG as a company has Indian options data**

from

> **IITB's LSEG subscription gives students access to the historical Indian options data.**

The second one needs to be verified.

For our project, I'd specifically ask the IITB library:

> **Does our LSEG Workspace subscription provide historical NSE equity/index options data at the contract level, including trade date, strike, expiry, call/put, price, volume and open interest?**

If yes, India becomes very viable.

---

3. Capitaline is interesting for Indian options

I initially wouldn't have put much weight on Capitaline, but after checking its current product information, there **is evidence that it covers derivatives**.

Capitaline explicitly says its market-data coverage includes securities, **derivatives**, and commodities traded on Indian stock markets. Capitaline +1

More specifically, its TradeCues product has a:

> **"Scans on Derivatives Data"**

section with open-interest and price analysis. TradeCues

So I would classify Capitaline as:

$$\boxed{\text{Indian derivatives: YES}}$$

but **not yet**

$$\boxed{\text{full historical option-chain research dataset: confirmed}}$$

The publicly available documentation doesn't establish that you can export a panel like:

```
Plain text
```

```
date
underlying
strike
expiry
CE/PE
bid
ask
close
volume
OI
IV
```

for thousands of historical contracts.

That is what we need to verify inside the IITB Capitaline account.

---

4. CMIE/Prowess: useful, but not for options

CMIE Prowess is actually very useful for the **portfolio side**.

It provides Indian:

- stock prices
- corporate actions
- daily total returns
- financial statements
- company fundamentals

and has time-series data going back to 1990. Prowessdx

So we could potentially have:

$$\text{CMIE} \rightarrow \text{Indian stocks} + \text{fundamentals}$$

but I would **not use CMIE as our options source**.

It can complement an options dataset.

For example:

$$\text{NSE/LSEG options} + \text{CMIE fundamentals}$$

could be useful.

5. Yahoo Finance

Yahoo is useful, but **not for the dataset we ultimately want**.

Yahoo does provide option chains for US securities, and there are open-source libraries/APIs that expose:

- expiration dates
- strikes
- calls
- puts

For example, the open-source Yahoo API implementations demonstrate retrieving option expiration dates and option contracts. GitHub

But there's a major issue:

We need historical option chains.

Suppose our training observation is:

$$t = \text{2021-06-01}$$

We need to know:

> What did the entire option surface look like **on 2021-06-01**?

Yahoo's ordinary historical-price infrastructure isn't designed to provide that complete historical option surface. Yahoo's own documentation is primarily about historical **price/dividend/split** data and notes that downloadable historical data availability varies by instrument. Yahoo Help

So:

$$\boxed{\text{Yahoo} \neq \text{our final research dataset}}$$

But it is perfectly fine for building the first prototype.

---

6. NSE is probably our strongest free/official Indian fallback

If we decide:

$$\boxed{\text{India}}$$

then NSE's own derivatives data becomes important.

We can potentially obtain historical contract-level F&O data containing things like:

$$\text{expiry}, K, \text{option type}, \text{price}, \text{volume}, OI$$

The difficulty is **how much historical data we can access conveniently and under what licensing/subscription arrangement**.

For a one-month academic project, I'd strongly prefer a clean institutional download over building a scraper.

---

7. What we actually need for the proposed research

This is the important part.

For the project:

> **Option-implied information →** Q **→** P **transformation → ML/regime model → portfolio**

we don't need every possible option.

Our minimum viable dataset is something like:

Option data

$$(t, S, K, T, \text{Call/Put}, P, V, OI)$$

where:

- t = observation date
- S = underlying price
- K = strike
- T = expiry
- P = option price
- V = volume
- OI = open interest

Then we can calculate:

$$IV$$

ourselves.

And from the cross-section:

$$C(K,T)$$

we can estimate the risk-neutral distribution.

Potentially:

$$q(S_T) \propto \frac{\partial^2 C(K,T)}{\partial K^2}$$

and extract:

$$E^Q[S_T]$$

$$Var^Q[S_T]$$

$$Skew^Q[S_T]$$

etc.

Then our ML/regime model attempts:

$$\boxed{Q\text{-information} \rightarrow P\text{-distribution}}$$

That's the interesting part.

---

8. Therefore, I'd rank your available options like this

If we want US

1. LSEG Workspace

⭐⭐⭐⭐⭐

This is the one to investigate at IITB.

2. Yahoo

⭐⭐

Only for prototyping, not final historical research.

3. Other open-source datasets

⭐

Potentially useful if we find a clean historical dataset, but I wouldn't design the project around one before verifying it.

If we want India

1. LSEG Workspace



if IITB's subscription includes NSE historical options.

2. Capitaline



Definitely has Indian derivatives coverage, but we need to verify the **granularity and historical export**. Capitaline +1

3. NSE



Potentially excellent official source, subject to historical-data accessibility/licensing.

4. CMIE



For underlying stocks/fundamentals, **not options**. Prowessdx

---

One thing I would do before choosing US vs India

Don't choose the market yet.

Go into **LSEG Workspace at IITB** and **Capitaline AWS** and see whether you can actually retrieve an historical option contract.

For example, try to answer:

> Can I obtain all NIFTY options on **15 January 2024**, including strike, expiry, CE/PE, price, volume and OI?

and:

> Can I obtain all SPY/SPX options on **15 January 2024**, with the same fields?

If either works, **we have our dataset**.

If you want, I can next give you a **very specific checklist of exactly what fields/data frequency/history we need to verify in LSEG, Capitaline and NSE**, so you can spend 10–15 minutes checking IITB access and come back with the result.

I read the abstract, intro, and conclusion of all the 4 papers

Good. Now we should **stop adding papers for the moment** and use those four to reconstruct the research landscape.

The next step I suggest is that you tell me, in your own words, **what you understood from each paper**—even if it is rough.

For each of the four, give me just:

1. **What problem does the paper solve?**
2. **What is their methodology?**
3. **What data do they use?**
4. **What is their main result?**
5. **What did you find confusing / unclear?**

You don't need to write a polished summary. Even something like:

> Paper 1: They take option IV and skewness → estimate something → put it into Markowitz. I'm not fully clear how they get expected returns.

is enough.

Then I can do something much more useful than another literature search:

We'll construct this map

$$\boxed{\text{Paper 1}} \quad \rightarrow \quad \text{what it establishes}$$

$$\boxed{\text{Paper 2}} \quad \rightarrow \quad \text{what it adds}$$

$$\boxed{\text{Paper 3}} \quad \rightarrow \quad \text{Q vs P / martingale framework}$$

$$\boxed{\text{Paper 4}} \quad \rightarrow \quad \text{where ML enters}$$

Then:

$$\boxed{\text{Existing literature}} \rightarrow \boxed{\text{what is already done}} \rightarrow \boxed{\text{what is missing}} \rightarrow \boxed{\text{our project}}$$

That last transition is the important one. We should **not assume** that adding Markov + ML + martingale automatically makes the project novel. We need to identify exactly what those four papers leave open and then formulate a contribution that can actually be completed in one month.

1. DeMiguel‡

- The paper tried to examine whether one can use option-implied information to improve the selection of mean-variance portfolios with a large number of stocks
- Use of option implied volatilities and correlations to improve out of-sample performance of mean-variance portfolios invested in only risky stocks. Next, examine the use of risk-premium corrected option-implied correlations to improve the performance of minimum-variance portfolios. Finally, to improve the out-of-sample performance of mean-variance portfolios they consider the use of option-implied volatility, risk premium for stochastic volatility, and option-implied skewness.
- S&P 500 stocks and options data and also the derived options data like IV and other stuffs
- They found option-implied volatilities can lead to a significant improvement in portfolio volatility. However, option-implied correlations are less useful in reducing portfolio volatility and also, forming portfolios using expected returns that exploit information in option-implied model-free skewness and implied volatility achieve a higher Sharpe ratio than portfolios that ignore option-implied information; even after adjusting for transaction costs. They also conclude that prices of stock options contain information that can be useful for improving the out-of-sample performance of portfolios.
- I didn’t understand the idea behind, how did they model the system exactly in detail to include all these options features in deciding the optimal portfolio weights and also how did they conclude the IV surface is very helpful, what metrics did they use to deduce that part.

2. Bianchia

- They suggest a portfolio selection framework based on option implied information and multivariate non-Gaussian models.
- They considered two models, which are a multivariate extension of the normal tempered stable (NTS) model and the generalized hyperbolic (GH) model and the connection between the historical measure P and the risk-neutral measure Q is given by the Esscher transform.
- They didn’t mention any particular data, as this one’s the theoretical implementation of the two models idea they have proposed.
- First, they proposed a multivariate option pricing framework based on heavy tails, negative skewness and asymmetric dependence. Second, they conduct a large scale empirical study based on a joint calibration-estimation of the univariate option surfaces and of the time series of log-returns has been proposed. Third, show how to use the proposed framework to evaluate portfolio risk measures. Finally, the multivariate non-Gaussian models proposed in this work together with the double calibration approach can be used to explore the interdependencies in financial markets, not only for solving portfolio allocation problems, but also as a statistical tool for financial stability purposes.
- I didn’t understand how they quantified the two models i.e., NTS and GH. And how did they incorporate it into option pricing.

3. Chernov

- The paper addresses the challenge of jointly estimating objective (historical) and risk-neutral probability measures to accurately price equity options while reconciling time-series dynamics with cross-sectional option prices.
- They propose a unified estimation framework using a Efficient Method of Moments (EMM) approach combined with stochastic volatility and jump-diffusion models to simultaneously fit underlying asset returns and option prices.

- Daily S&P 500 index returns along with S&P 500 index call option prices across various maturities and strike prices.
- Incorporating both stochastic volatility and price jumps is essential for joint consistency between objective time-series return models and risk-neutral option pricing models, significantly reducing option pricing errors.
- The precise parameterization of the risk premium specifications linking the physical and risk-neutral measures and the empirical identification of volatility jump sizes were somewhat complex and hard to follow.

4. Bali

- The paper addresses the challenge of predicting individual equity option returns in a high-dimensional setup, where traditional linear models struggle due to complex nonlinearities and interaction effects among numerous predictive features.
- They implement machine learning methods (such as neural networks and tree-based models) combining options-specific characteristics, underlying stock signals, and macroeconomic variables to forecast out-of-sample option returns and construct long-short portfolios.
- Large-scale US equity options dataset (OptionMetrics) alongside underlying stock characteristics (CRSP/COMPUSTAT) and macroeconomic indicators.
- Machine learning models significantly outperform traditional benchmark linear models in predicting option returns out-of-sample, generating economically and statistically significant Sharpe ratios even after accounting for transaction costs and market friction.
- It was not entirely clear which specific categories of features (e.g., options-level vs. stock-level characteristics) contributed most to the predictive performance during market stress periods versus normal market conditions.

Your understanding is broadly correct. More importantly, your four summaries reveal that the papers are **not four independent ideas**—they form a fairly coherent chain:

$$\boxed{\text{Option surface} \to Q\text{-distribution} \to P\text{-distribution} \to \text{forecast} \to \text{portfolio}}$$

But there are some important gaps in your current understanding. Before deciding our contribution, I would resolve those gaps.

1. DeMiguel — this should become our portfolio baseline

Your interpretation is correct.

The important thing you missed is **how option information enters the portfolio problem**.

They aren't simply saying:

> "IV is high → increase/decrease the stock weight."

Instead, think of the pipeline as:

$$\text{Option prices} \to \text{option-implied quantities} \to \text{estimates of moments} \to (\hat{\mu}, \hat{\Sigma}) \to \text{Markowitz} \to w^*$$

The fundamental portfolio problem remains something like

$$\max_{w} \quad w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w$$

The research question is:

> Can option information produce **better estimates of μ and/or Σ** than conventional historical estimates?

That's why their evaluation is ultimately based on **out-of-sample portfolio performance**, not simply whether the IV surface "looks informative."

The relevant metrics are things such as:

- realized portfolio volatility
- Sharpe ratio
- portfolio returns
- turnover
- transaction costs

- statistical significance of performance differences

So when you later read the methodology, don't think:

> "How did they use IV as a trading signal?"

Think:

> **"Which component of the Markowitz inputs did they replace/improve using option information?"**

That is the key.

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2. Bianchi — your biggest conceptual gap

Your question:

> "How did they quantify NTS and GH and incorporate it into option pricing?"

is exactly the right question.

But don't start by trying to understand all the mathematics of NTS/GH.

At the conceptual level, they are doing:

$$\text{historical returns} \rightarrow \text{estimate distribution under } P$$

and

$$\text{option prices} \rightarrow \text{information about distribution under } Q$$

Then they impose a particular relationship:

$$\boxed{P \longleftrightarrow Q}$$

through the **Esscher transform**.

The NTS/GH models give them a flexible distribution that can represent:

- fat tails
- skewness
- dependence between assets

rather than assuming simple Gaussian returns.

So their contribution is basically:

> "Let's build a sufficiently flexible multivariate distribution, make it consistent with option prices under Q, connect it to historical returns under P, and then use the resulting distribution for portfolio risk/selection."

You don't need to reproduce NTS/GH yet.

For our project, the lesson is:

$$\boxed{\text{Option prices do not directly give us the investor's expected return.}}$$

There is a **measure transformation/model connecting Q to P**.

That's going to be extremely important if we pursue your idea.

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3. Chernov — this explains why the Bianchi idea is necessary

Your summary is accurate.

Chernov is effectively asking:

> Can we construct a model that is simultaneously consistent with historical returns and observed option prices?

Their architecture is approximately:

$$\text{Historical S\&P returns} \rightarrow P\text{-dynamics}$$

while simultaneously:

$$\text{S\&P option prices} \rightarrow Q\text{-pricing}$$

and the model must connect the two.

The important takeaway for us is **not EMM itself**.

It is:

$$\boxed{\text{A model can fit historical returns beautifully but price options badly.}}$$

and conversely:

$$\boxed{\text{A model can fit option prices beautifully but describe physical returns poorly.}}$$

Chernov's joint estimation attempts to prevent those two worlds from becoming disconnected.

This is directly relevant to our project.

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4. Bali — this tells us how we can use ML

You understood this one well.

The important thing to extract is the research methodology:

$$X_t = [\text{option characteristics}, \text{stock characteristics}, \text{macro variables}]$$

then:

$$\hat{y}_{t+1} = f(X_t)$$

and evaluate strictly **out-of-sample**.

This is where your existing ML experience becomes useful.

You don't need Bali to teach you Random Forest/XGBoost/NN.

The useful part is:

> **How do we turn a huge option dataset into economically meaningful predictive features and then evaluate whether those predictions actually improve a portfolio?**

Now look at what happens when we put the papers together

This is where I think the project becomes interesting.

Existing Paper 1

DeMiguel

$$\text{Option information} \rightarrow \text{portfolio}$$

Existing Paper 2

Bianchi

$$P \leftrightarrow Q \rightarrow \text{portfolio risk}$$

Existing Paper 3

Chernov

$$\text{return dynamics} + \text{option prices} \rightarrow \text{joint } P/Q \text{ model}$$

Existing Paper 4

Bali

$$\text{high-dimensional option/stock/macro features} \rightarrow ML \rightarrow \text{future option returns}$$

And here's the interesting gap

The papers give us four pieces:

$$\boxed{Q}$$

$$\boxed{P}$$

$$\boxed{Q \leftrightarrow P}$$

$$\boxed{ML}$$

but they don't simply combine them into the exact architecture we're considering.

Our potential research question could therefore become:

> **Can machine learning exploit option-implied risk-neutral information to estimate economically useful physical-measure return distributions, and does incorporating these forecasts into portfolio optimization improve out-of-sample performance?**

And we can introduce your **Markov/regime-switching component**:

$$Z_t = \text{latent market regime}$$

such that:

$$P(R_{t+1} \mid X_t, Z_t)$$

rather than assuming one stationary relationship between option-implied Q-information and future P-returns.

That gives:

$$\boxed{\text{Options} \rightarrow Q\text{-features} \rightarrow \text{Regime} \rightarrow ML \rightarrow P\text{-forecast} \rightarrow \text{Portfolio}}$$

But I would make one correction to our earlier idea

You originally said:

> "use ML algos to predict option pricing and use that to get an expected stock price under no-arbitrage"

I **would not do that**.

It creates an unnecessary intermediate problem:

$$ML \rightarrow \widehat{\text{Option Price}} \rightarrow Q \rightarrow P$$

Why predict option prices if we already have observed option prices?

Instead:

$$\boxed{\text{Observed option prices} \rightarrow Q\text{-information}}$$

and then:

$$\boxed{Q\text{-information} \rightarrow ML/regimemodel \rightarrow P}$$

The ML problem should be the **Q-to-P/investment-information problem**, not merely option-price prediction.

That is also much more aligned with your four papers.

The project now has three distinct layers

Layer 1 — Financial engineering

From option prices:

$$C(K,T), P(K,T)$$

derive:

$$IV, \quad \text{skew}, \quad \text{term structure}, \quad Q\text{-moments}, \ldots$$

This is the **IE612** component.

Layer 2 — Statistical/ML

Use those variables to estimate:

$$\hat{\mu}_{t+1}$$

and potentially:

$$\hat{\sigma}_{t+1}, \quad \hat{\text{skew}}_{t+1}$$

conditioned on regime:

$$Z_t.$$

This is your **ML component**.

Layer 3 — Portfolio

Use:

$$(\hat{\mu}, \hat{\Sigma})$$

in:

$$w^* = \arg\max_w \left[w^\top \hat{\mu} - \frac{\lambda}{2} w^\top \hat{\Sigma} w \right]$$

and compare against:

- historical Markowitz
- option-implied baseline
- ML without options
- proposed option + regime + ML model

This is the **economic validation**.

One thing I would NOT add yet

Don't add all of these simultaneously:

- HMM
- Markov chain
- NTS
- GH
- Esscher
- stochastic volatility
- jumps

- Black-Scholes
- XGBoost
- Random Forest
- neural network
- LSTM
- RL

That would turn into a **kitchen-sink project** and become impossible to explain.

Instead, we need to determine:

$$\boxed{\text{What is the minimum new mechanism that produces a defensible contribution?}}$$

My current candidate is:

> **Regime-conditioned ML estimation of the physical return distribution using option-implied risk-neutral information, followed by portfolio optimization.**

But **we haven't established that this is actually novel yet**.

That's the next research task I would do.

We should now go **one level deeper into the methodology of these four papers**, specifically extracting:

1. **What exactly goes in**
2. **What exactly comes out**
3. **Mathematical transformation**
4. **Data required**
5. **Portfolio construction**
6. **Evaluation metrics**
7. **What each paper does NOT do**

Once we have that table, we can identify the missing cell(s). That missing cell is where our one-month project should live.

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I think this is good to go, We should keep it final and the extra recommended paper which is a gap here has to be added

Yes. We can **freeze the literature set at 5 papers** for the initial project-design phase.

One correction in terminology: the extra paper, **Ulrich, Walther, Rothfuss & Ferreira, "Forward-looking P" (2019)**, is **not itself the gap**. It is important because it closes a major part of the apparent gap: it already uses option-implied risk-neutral information with a neural network to estimate the physical return distribution. Pasted text

So the final reading set is:

#	Paper	Role in our project
1	**DeMiguel et al. (2013), Improving Portfolio Selection Using Option-Implied Volatility and Skewness**	Option information → portfolio optimization
2	**Bianchi & Tassinari (2020), Forward-looking Portfolio Selection with Multivariate Non-Gaussian Models**	P–Q connection + portfolio risk
3	**Chernov & Ghysels (2000), A Study Towards a Unified Approach to the Joint Estimation of Objective and Risk Neutral Measures**	Joint P/Q modelling
4	**Bali et al. (2023), Option Return Predictability with Machine Learning and Big Data**	ML + option information + rigorous OOS prediction
5	**Ulrich et al. (2019), Forward-looking P**	**Critical prior art for option-implied Q-information → physical P-density using neural networks**

The fifth paper is particularly important because its architecture is already close to:

$$\text{Options} \rightarrow Q\text{-moments} \rightarrow \text{neural density model} \rightarrow \hat{p}_t(r)$$

using a mixture-density network. Pasted text

That changes how we should formulate our eventual contribution. We **cannot** claim novelty for merely:

> "Use ML to transform option-implied information into physical expected returns."

Likewise, adding an HMM by itself is insufficient, because option-implied skewness regimes combined with portfolio optimization have already been studied. Pasted text

The potentially defensible direction is narrower:

$$\boxed{\text{persistent latent regime dependence}}$$

in the relationship

$$Q\text{-information} \rightarrow P\text{-density}$$

and testing whether that persistence gives incremental value over a sufficiently flexible **non-regime** model and ordinary time-varying calibration. The audit explicitly identifies this as the candidate extension, while stressing that its novelty is still provisional. Pasted text

So I would now consider these **5 papers our core literature pack**. We don't need another batch of papers yet. The next step is to extract the actual mathematical/model architecture from these five and turn it into a concrete one-month implementation with a clearly defined baseline, proposed model, and novelty test.